

23ELC212 Machine Learning



Health Monitoring and Fault Detection in Naval Vessel Systems.

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I. Introduction

- **Propulsion systems** and **IGBTs** undergo progressive degradation due to mechanical, thermal, and electrical stresses in a **Naval/Marine Vessel**.
- Mechanical wear in propulsion affects thrust efficiency and system stability.
- Electrical and thermal stresses in IGBTs cause parameter drift, increased losses, and switching degradation.
- Degradation in one subsystem can influence the performance of the other.
- Most existing approaches monitor propulsion and power electronics independently, without considering their interdependency.
- Lack of integrated health assessment limits accurate system-level reliability prediction.



II. Objective

- The objective of this work is to develop a machine learning–based health monitoring and fault detection framework for naval vessel propulsion and its power systems.
- To monitor both **mechanical parameters** (Lever position ,Ship speed ,Gas Turbine ,shaft torque ,rate of revolutions ,Gas Generator rate of revolutions etc.) and **electrical parameters** (Steady state, transient state(switching) , Thermal features of IGBTs) and compute overall health.
- To estimate the Remaining Useful Life (RUL) of the IGBT module using **Regression techniques**
- To model and predict the degradation state of the ship propulsion system using **Regression techniques**
- To compute a unified Ship Health Index (SHI) representing the combined health state of mechanical and electrical subsystems using **Fuzzy Logic**.

III. Literature Survey



Ref. No.	Author	Title, Year	Remarks
[1]	Francesca Cipollini, Luca Oneto, Andrea Coraddu, Alan John Murphy, Davide Anguita	Condition-Based Maintenance of Naval Propulsion Systems with Supervised Data Analysis, 2018	• Proposes a data-driven condition-based maintenance (CBM) framework for naval propulsion systems. • Demonstrates that complex ship propulsion behavior cannot be accurately modeled using only prior physical knowledge and motivates the use of machine learning. • Models degradation of four components: gas turbine, compressor, propeller, and hull.-Novelty • Formulates the problem as a multi-output regression task to estimate component decay coefficients from operational sensor data. 15 combination prediction problems. • Evaluates multiple ML models including deep neural networks, Gaussian processes, SVR, random forests, and k-NN. • Concludes that deep neural networks provide the best predictive performance
[2]	Dishita Sood, Nonita Sharma, Manoj Soni, Manik Rakhra	ML-based Prediction of the Decay Coefficient of a Naval Propulsion System's Turbine, 2024	• Focuses specifically on predicting the turbine decay coefficient of a naval propulsion system using machine learning regression techniques. • Uses the UCI naval propulsion dataset • Performs extensive feature engineering to create physics-informed variables including total torque, propulsion power, fuel efficiency, and system efficiency. • Compares multiple regression algorithms including Linear Regression, Decision Trees, Random Forest, KNN, ANN, and Support Vector Machines. • Shows that hyperparameter-tuned Support Vector Machine with RBF kernel achieves the best performance ($R^2 \approx 99.22\%$). • Demonstrates that combining domain knowledge with machine learning significantly improves degradation prediction accuracy. • Confirms the suitability of SVR-based models for predictive maintenance of ship propulsion turbines.

III. Literature Survey



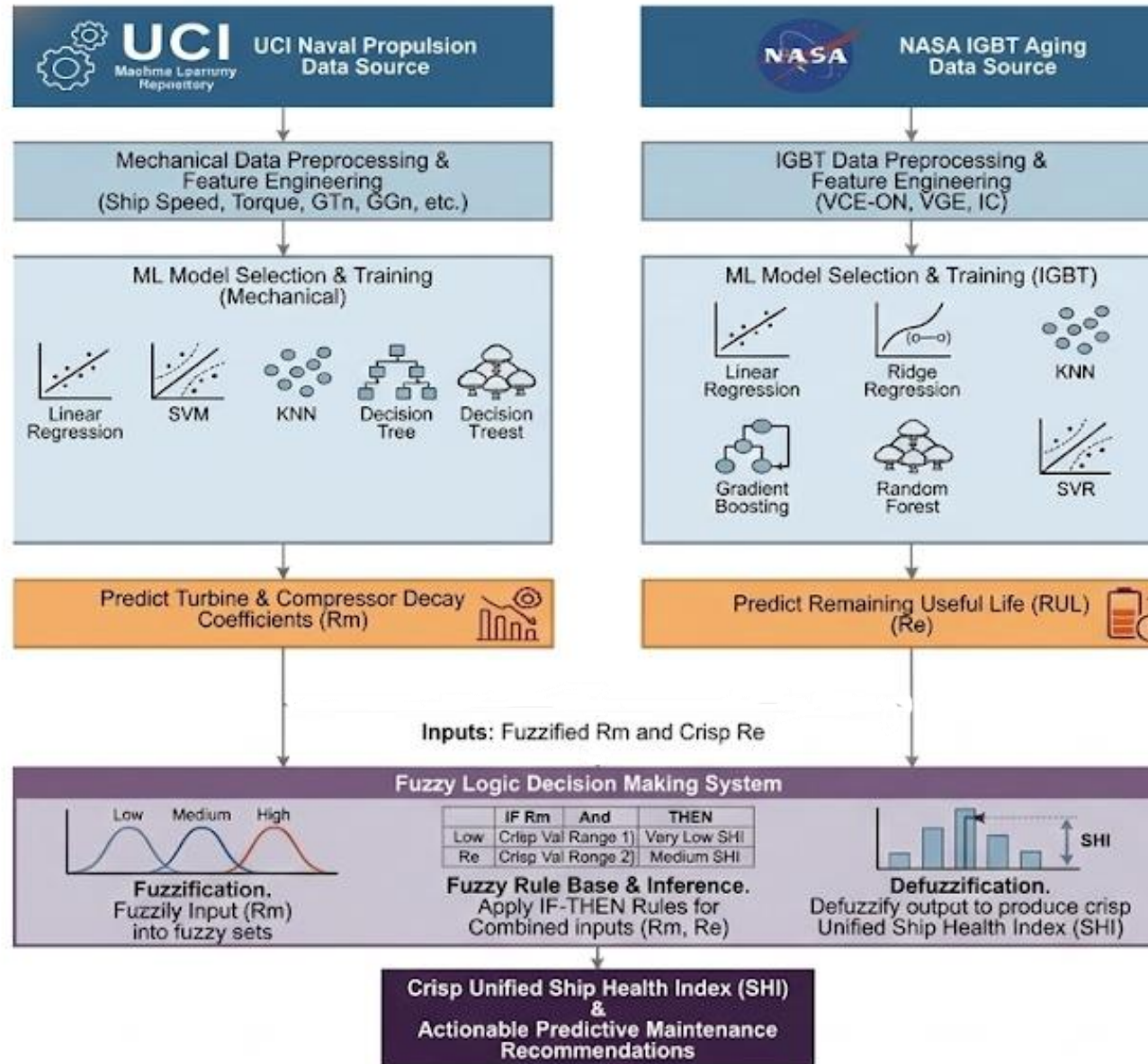
Ref. No.	Author	Title, Year	Remarks
[3]	Nishad Patil, Jose Celaya, Diganta Das, Kai Goebel, Michael Pecht	Precursor Parameter Identification for Insulated Gate Bipolar Transistor (IGBT) Prognostics, IEEE TRANSACTIONS ON RELIABILITY, VOL. 58, NO. 2, JUNE 2009	- Investigates precursor parameter identification for IGBT prognostics using accelerated thermal overstress aging experiments. - Identifies Collector–Emitter ON voltage (VCE(ON)) reduction as a primary electrical indicator of die-attach degradation. - Observes increase in turn-off delay time as the device approaches latch-up failure. - Correlates electrical parameter drift with physical degradation using Scanning Acoustic Microscopy (SAM) analysis. - Concludes that precursor-based electrical monitoring enables early degradation detection and supports Remaining Useful Life (RUL) estimation of IGBTs.
[4]	Theodoros Dimitriou, Emmanouil Skondras, Christos Hitiris, Cleopatra Gkola, Ioannis S. Papapanagiotou, Dimitrios J. Vergados, Stavros I. Papapanagiotou, Stratos Koumantakis, Angelo Michals, and Dimitrios D. Vergados	Supervised Machine Learning Algorithms for the Analysis of Ship Engine Data, pp 97-109, 20 May 2024.	- The study presents a comparative evaluation of supervised machine learning algorithms including Linear Regression, Ridge Regression, Decision Tree, and Ensemble methods using the CBM ship engine dataset. - The CBM dataset contains 11,934 records with 16 features, out of which 14 relevant features were used after eliminating constant parameters to improve analysis accuracy. - Ensemble-based models such as Bagging, Random Forest, and Extra Trees demonstrated superior performance in predicting gas turbine compressor decay factors due to reduced bias and variance. - Decision Tree and k-Nearest Neighbor models showed better prediction capability than linear regression models, indicating the presence of non-linear relationships in ship engine data. - Linear Regression and Ridge Regression exhibited comparatively lower performance, suggesting that engine decay parameters do not follow strictly linear trends. - Artificial Neural Network showed the least prediction accuracy, highlighting the need for further architecture tuning and parameter optimization for effective implementation.

III. Literature Survey



Ref. No.	Author	Title, Year	Remarks
[5]	G. Sonnenfeld, K. Goebel, J. Celaya	An Agile Accelerated Aging, Characterization and Scenario Simulation System for Gate Controlled Power Transistors, 2008	•Developed an accelerated aging platform capable of inducing thermal cycling, electrical overstress, and hot carrier injection in IGBTs. •Thermal stress accelerates die-attach solder fatigue and bond wire degradation, increasing thermal resistance (R_{th}). •Electrical overstress causes gate oxide degradation and threshold voltage shifts. •Monitored steady-state and transient parameters including VCE, VGE, switching transients, and current waveforms. •Observed that degradation leads to increased conduction losses, switching delay, and voltage drop variations. •These electrical parameter variations are used as health indicators for prognostics and RUL estimation.
[6]	R. V. Prathiba, M. Saravanan, P. W. David, and H. Thangaraj	Machine Learning-Based Direct Estimation of Remaining Useful Life (RUL) of IGBT Using Multi-Precursor Prognostics, vol 107, 2025	•Used NASA accelerated aging dataset to analyze degradation trends in VCE(ON), IC, and VGE. •Extracted time-domain features such as mean, RMS, variance, skewness, and trend slope to represent degradation progression. •Demonstrated that single-precursor models are insufficient due to nonlinear aging behavior. •Applied ML models with feature ranking and hyperparameter tuning for improved RUL estimation. •Showed that combining multiple degradation indicators enhances robustness and prediction accuracy. •Established a data-driven framework for estimating IGBT health state under progressive aging.

IV. Block Diagram



V. Methodology



Mechanical Subsystem

- Naval Propulsion Plant dataset used, captures propulsion behavior under varying operational and sea conditions
- 8 redundant/duplicate columns dropped (Vessel_Speed, CPP_T_port, Q_stdb, rpm_stbd, Kt_port, rps_prop_port, Q_prop_stbd, Kkq)
- Statistical features extracted from degradation signals, data normalized for multi-output regression targeting 4 decay coefficients — Kkt, Khull, KMcompr, KMturb
- 12 regression algorithms evaluated: Linear, Polynomial, MLR, Lasso, Ridge, Elastic Net, SVR, KNN, Decision Tree, Random Forest, AdaBoost, Gradient Boosting
- Hyperparameters tuned via grid search — C, epsilon, gamma (SVR); n_estimators, max_depth(ensembles); alpha, l1_ratio (regularization)
- All models confirmed using R^2 , RMSE and MAE; best model selected for decay coefficient prediction

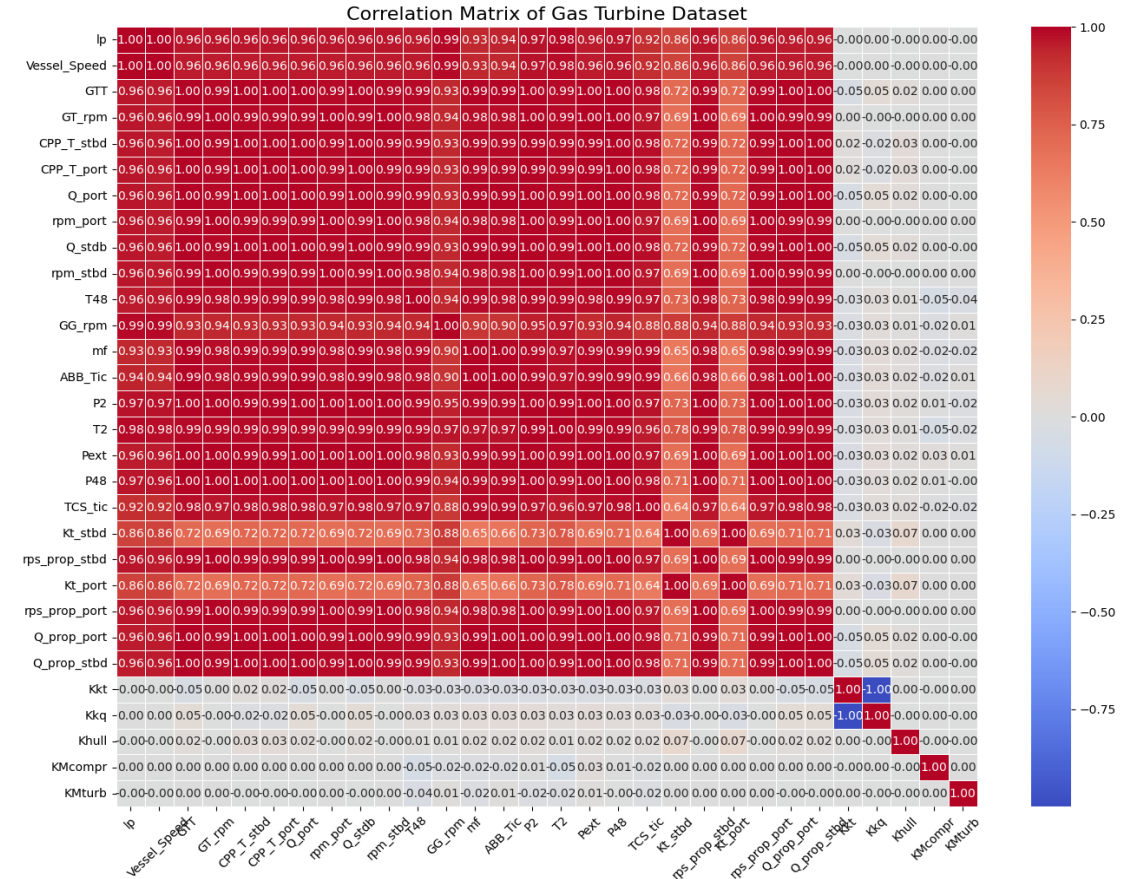


Fig. 5.1 Heatmap for ships propulsion system dataset

V. Methodology



IGBT RUL Prediction

- 41,752 rows, 418 thermal cycles, RUL 0–100%
- Time proxies excluded ($r > 0.99$ with RUL) — 12 physical precursor features retained
- Transient V_{ce_max} identified as dominant signal ($r = 0.51$), validated by Lasso (only survivor at $\alpha = 10$)
- Group-based split on Transient_Block_ID — prevents leakage across aging cycles
- StandardScaler() applied to distance/kernel models only (SVR, KNN, linear); tree-based models unscaled
- 12 regression algorithms trained with grid search hyperparameter tuning per model
- Evaluated on R^2 , RMSE, MAE — Gradient Boosting best at Test $R^2 = 0.956$
- Linear models yielded negative R^2 , confirming nonlinear degradation behavior.

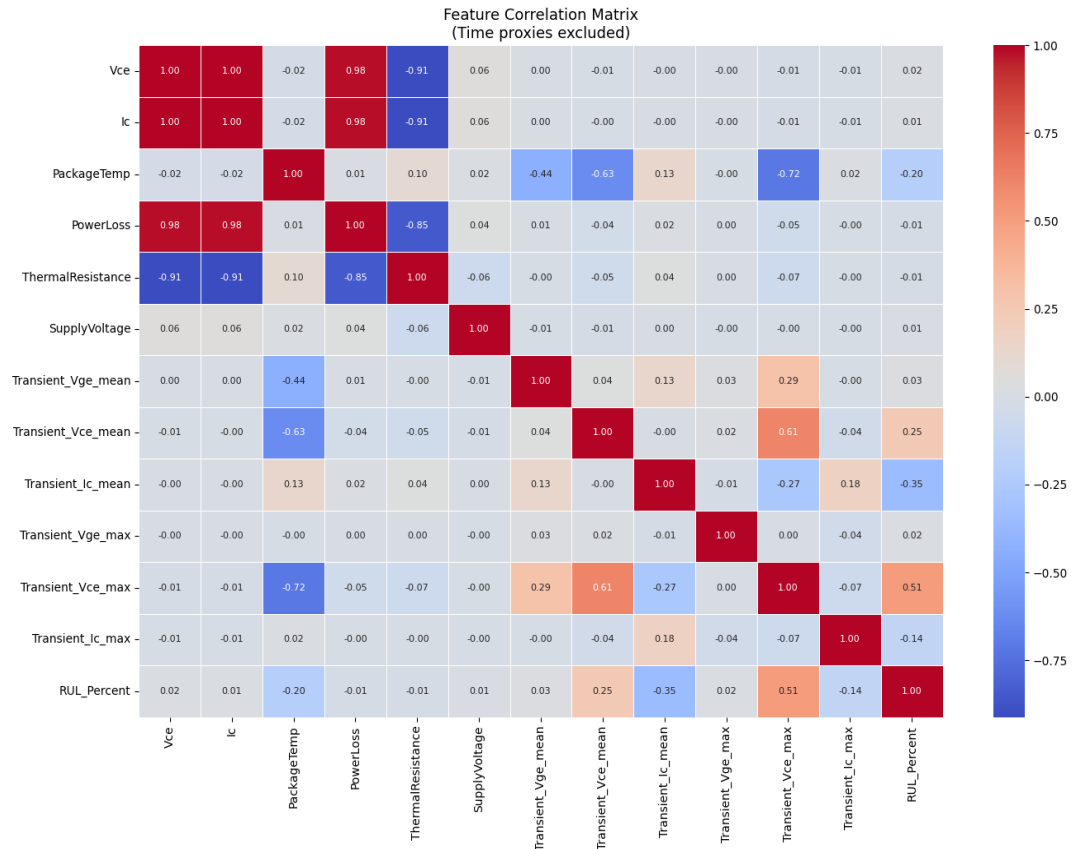


Fig. 5.2 Heatmap of GBT RUL dataset

V. Methodology



Fuzzy Logic Health Assessment – WDA + RUL Fusion

- Two inputs fed to the FIS: Weighted Decay Aggregate (WDA)[range: 0.930–1.000]and IGBT Remaining Useful Life (RUL %)[range: 0–100%]
- WDA is computed by fusing four decay coefficients:

$$WDA = 0.25 \cdot K_{kt} + 0.10 \cdot (2 - K_{hull}) + 0.35 \cdot K_{Mcompr} + 0.30 \cdot K_{Mturb}$$
- Three linguistic labels defined for each variable Poor, Average, and Good —Membership boundaries set directly from your data ranges:
WDA Poor: 0.930–0.955 | Average: 0.948–0.978 | Good: 0.972–1.000
RUL Poor: 0–35% | Average: 20–80% | Good: 65–100%
- A 3×3 rule matrix (9if-then rules) covers all WDA–RUL combinations, following: *IF WDA is X AND RUL is Y THEN Health is Z*
- Centroid defuzzification converts aggregated output into a crisp health score $\in [0, 1]$
- Final health label assigned by maximum membership:
Poor[0.0–0.4] | Average[0.4–0.7] | Good [0.7–1.0]

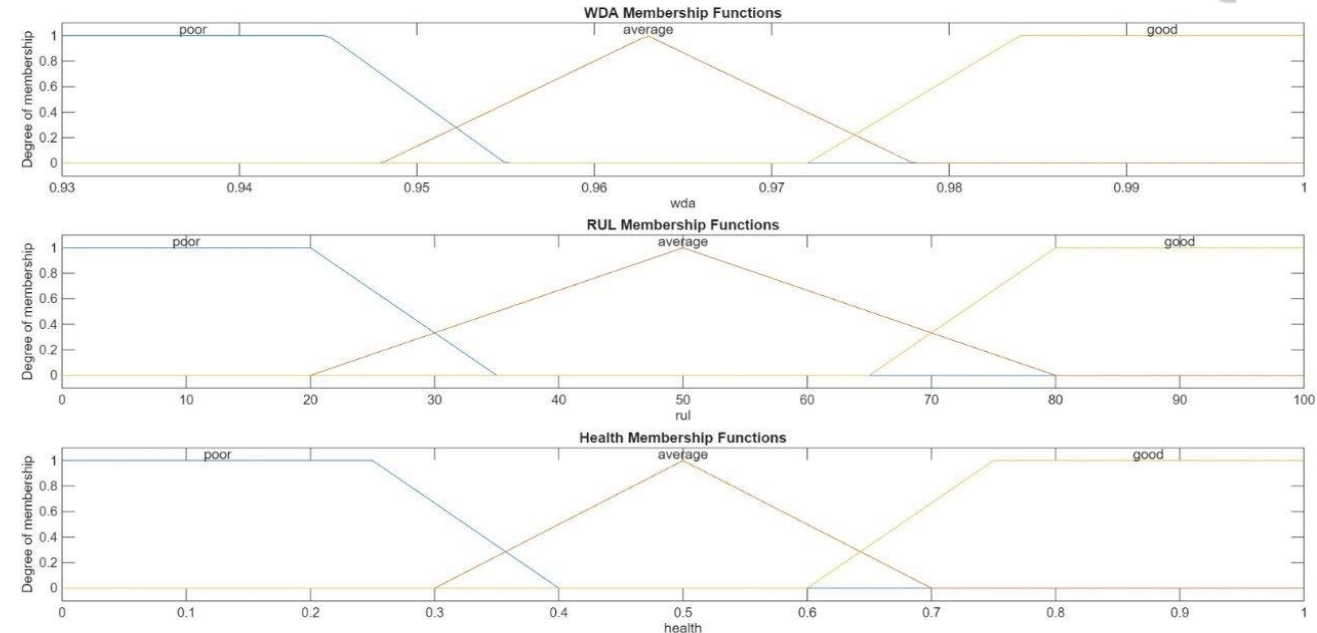


Fig. 5.3 Membership function vs rul , health and K (weighted decay coefficient)

WDA \ RUL	Poor	Average	Good
Poor	Poor	Poor	Average
Average	Poor	Average	Average
Good	Average	Good	Good

Fig. 5.3 Rule table of FUZZY logic



VII. Individual Contribution

Name	Contribution
Dharineesh MV	Mechanical system and fuzzy logic
Harshita JA	IGBT and Mechanical system and documentation
Haritha SK	IGBT and MatLab fuzzy logic

VIII. Results and Discussion



Fig. 8.1 – Model Performance Comparison: 12-Model RMSE comparison

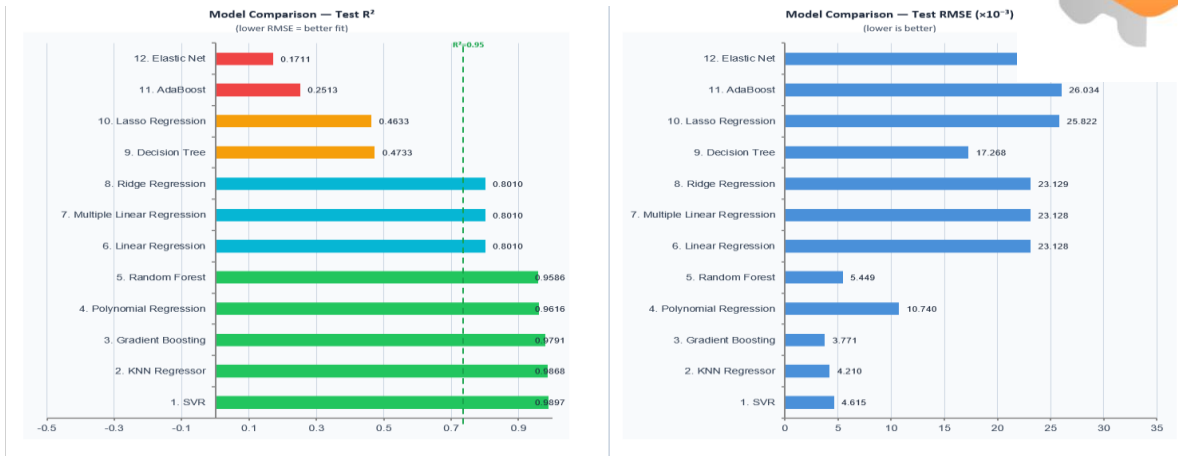


Fig. 8.2 – Model Performance Comparison: 12-Model Benchmarking for Ship Engine Decay Coefficient Prediction

Rank	Model	Best Config	Train R^2	Test R^2	Train MAE	Test MAE	Train RMSE	Test RMSE
1	SVR	<i>RBF, C=10, $\epsilon=0.001$</i>	0.9897	0.9897	0.001780	0.001778	0.004614	0.004615
2	KNN Regressor	<i>k=5, Euclidean</i>	0.9917	0.9868	0.001139	0.001482	0.003336	0.004210
3	Gradient Boosting	<i>200 trees, lr=0.1, depth=5</i>	0.9796	0.9791	0.002379	0.002402	0.003730	0.003771
4	Polynomial Regression	<i>Degree = 2</i>	0.9617	0.9616	0.005006	0.005020	0.010713	0.010740
5	Random Forest	<i>50 trees, depth=20</i>	0.9613	0.9586	0.001818	0.002030	0.005084	0.005449
6	Linear Regression	<i>Default</i>	0.8008	0.8010	0.011873	0.011872	0.023133	0.023128
7	Multiple Linear Regression	<i>Default</i>	0.8008	0.8010	0.011873	0.011872	0.023133	0.023128
8	Ridge Regression	<i>$\alpha = 0.0001$</i>	0.8008	0.8010	0.011874	0.011873	0.023133	0.023129
9	Decision Tree	<i>depth=20, min_split=2, min_leaf=1</i>	0.9546	0.9509	0.001488	0.001701	0.004733	0.005016
10	Lasso Regression	<i>$\alpha = 0.0001$</i>	0.4625	0.4633	0.016249	0.016246	0.025821	0.025822
11	AdaBoost	<i>200 estimators, lr=0.1</i>	0.2517	0.2513	0.018548	0.018575	0.025997	0.026034
12	Elastic Net	<i>$\alpha = 0.001, l_1_ratio = 0.3$</i>	0.1702	0.1711	0.021228	0.021220	0.031031	0.031036

- 12 models benchmarked across all 4 decay targets (Kkt, Khull, KMcompr, KMturb)
- SVR (RBF, C=10, $\epsilon=0.001$) — best Test $R^2 = 0.9897$, RMSE = 0.004615
- KNN (k=5, Euclidean) — second at $R^2 = 0.9868$
- Gradient Boosting (200 trees, lr=0.1) — third at $R^2 = 0.9791$
- Top 3 all exceed $R^2 = 0.97$ with near-identical train/test metrics — no overfitting
- Linear, MLR, Ridge converged at $R^2 \approx 0.80$ — confirms nonlinear decay behaviour
- Decision Tree, Lasso, AdaBoost, Elastic Net below $R^2 = 0.50$ — severe underfitting
- **Best model: SVR selected for decay coefficient prediction**



VIII. Results and Discussion

Results – Experiment 2: IGBT RUL Prediction (Remaining Useful Life %)

- IGBT degradation features extracted from capacitance, leakage current, and threshold voltage profiles
- Gradient Boosting again ranked first with Test $R^2 = 0.9561$, RMSE = 6.87
- Random Forest achieved Test $R^2 = 0.9155$, selected as the secondary model due to its stability and interpretability
- SVR (RBF Kernel) performed competitively at $R^2 = 0.8692$ after distance-based feature scaling
- All nine physically-informed models outperformed naive linear baselines by a significant margin
- Results confirm that IGBT RUL can be reliably estimated from physical measurement features alone, without time-series preprocessing

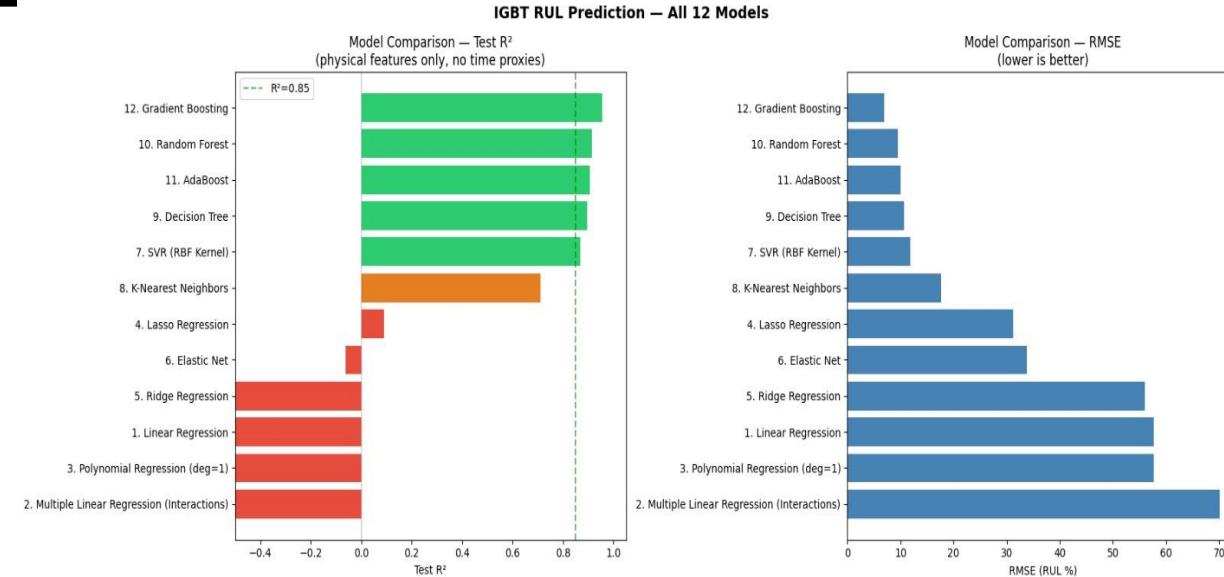


Fig. 8.3 – IGBT RUL Prediction: Test R^2 Comparison Across 12 Models

Rank	Model	Train R^2	Test R^2	RMSE	MAE	Scaling
1	Gradient Boosting	0.9998	0.9561	6.8734	5.0175	No (threshold)
2	Random Forest	1.0000	0.9155	9.5326	7.3447	No (threshold)
3	AdaBoost	0.8941	0.9063	10.0423	8.6197	No (threshold)
4	Decision Tree	1.0000	0.8951	10.6220	6.7617	No (threshold)
5	SVR (RBF Kernel)	0.9129	0.8692	11.8607	8.7542	Yes (distance)
6	K-Nearest Neighbors	1.0000	0.7112	17.6251	12.7639	Yes (distance)
7	Lasso Regression	0.1018	0.0913	31.2649	28.5654	Yes
8	Elastic Net	0.4387	-0.0631	33.8182	23.4654	Yes
9	Ridge Regression	0.5205	-1.9222	56.0673	23.4549	Yes
10	Linear Regression	0.5222	-2.0978	57.7278	23.5200	Yes
11	Polynomial Regression (deg=1)	0.5222	-2.0978	57.7278	23.5200	Yes
12	Multiple Linear Regression (Interactions)	0.5222	-2.0978	57.7278	23.5200	Yes

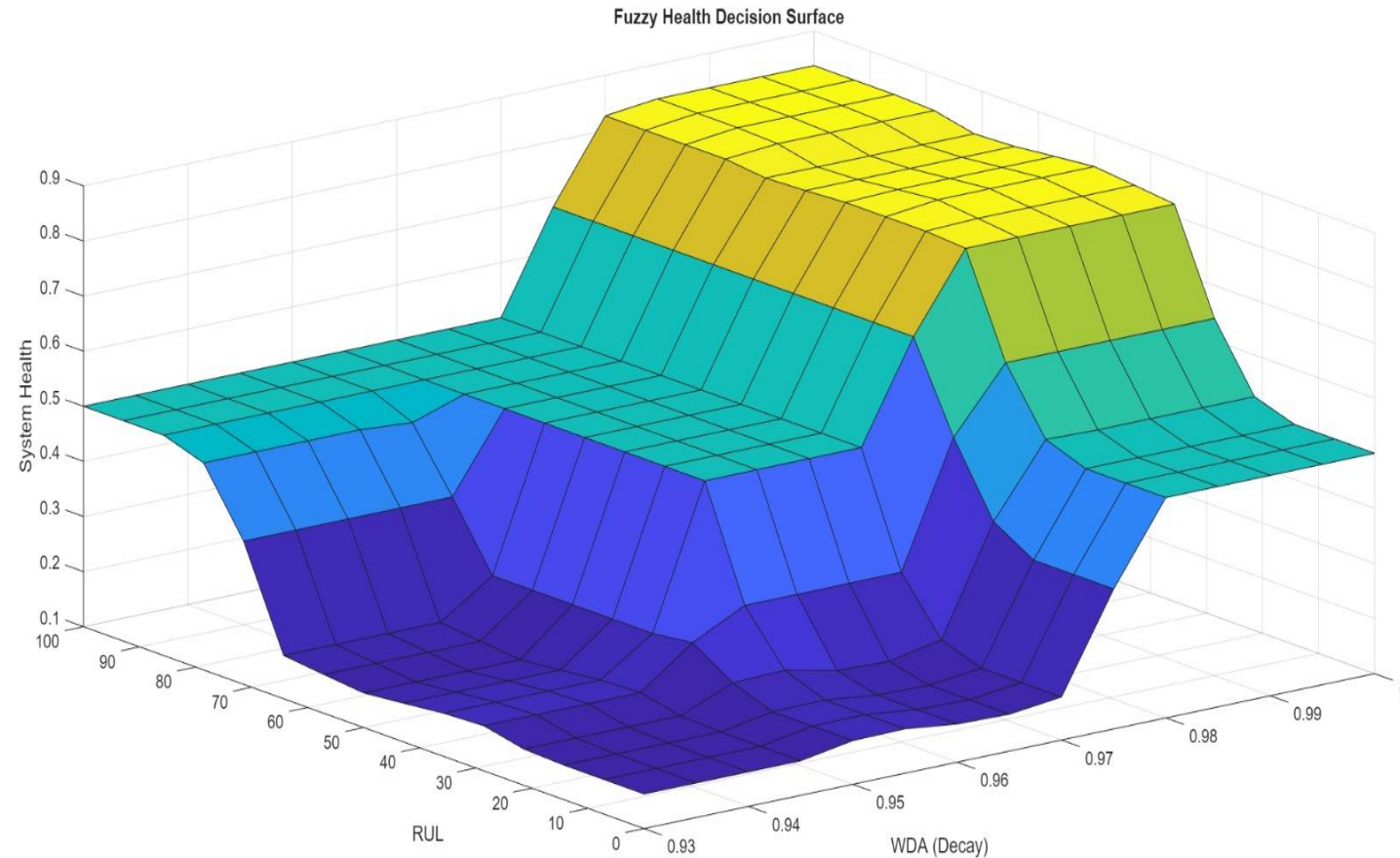
Test $R^2 \geq 0.90$ Test R^2 0–0.75 Test $R^2 < 0$ (poor fit)

Fig. 8.4 – IGBT RUL Prediction: RMSE Comparison Across 12 models

VIII. Results and Discussion



- A fuzzy system was built using two inputs — WDA and RUL — to predict overall System Health
- Three categories were defined for each input and output: Poor, Average, and Good
- WDA ranges from 0.93 to 1.0 | RUL ranges from 0 to 100% | Health output ranges from 0 to 1
- When $RUL > 70\%$ and $WDA > 0.97$, System Health is above 0.85 — the system is in Good condition
- When RUL drops below 30%, health is low regardless of WDA — the IGBT is near end of life
- The 3D surface shows smooth and consistent results with no sudden jumps or contradictions
- The output can be directly used to decide when maintenance is needed



IX. Conclusions



Summary of Main Findings

- SVR and Gradient Boosting consistently ranked as the top models across both propulsion and IGBT experiments, confirming the superiority of kernel-based and ensemble methods for degradation prediction
- Linear models failed across both datasets with negative or near-zero R^2 , proving that naval system degradation is inherently nonlinear
- IGBT health dominates overall system health — even a healthy propulsion system cannot compensate for a failing IGBT component
- The fuzzy inference system successfully unified mechanical and electrical health into a single interpretable score between 0 and 1

Key Contributions

- This study provides a comprehensive benchmark of 12 ML models across two distinct degradation prediction tasks — propulsion decay and IGBT RUL — on real sensor datasets
- It demonstrates that a lightweight fuzzy logic system can effectively fuse outputs from two independent ML pipelines into a unified vessel health index
- It establishes that model choice should depend on the task — SVR for smooth continuous degradation, Gradient Boosting for nonlinear RUL estimation
- The framework enables data-driven maintenance decisions for naval vessels using only sensor measurements, without physical inspection



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- Nishad Patil,Jose Celaya,Diganta Das,Kai Goebel,Michael Pecht — "Precursor Parameter Identification for InsulatedGate Bipolar Transistor (IGBT) Prognostics" , IEEE , Vol. 58, NO. 2, June 2009



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- R. V. Prathiba, M. Saravanan, P. W. David, and H. Thangaraj, "*Machine Learning-Based Direct Estimation of Remaining Useful Life (RUL) of IGBT Using Multi-Precursor Prognostics* " , Vol 107 , pp 3435-3449,2025.



References

Dataset:

- **OpenML – Dataset #41012**
- <https://www.openml.org/search?type=data&sort=runs&id=41012&status=active>
- **NASA – IGBT Accelerated Aging Dataset**
- <https://data.nasa.gov/dataset/insulated-gate-bipolar-transistor-igbt-accelerated-aging>